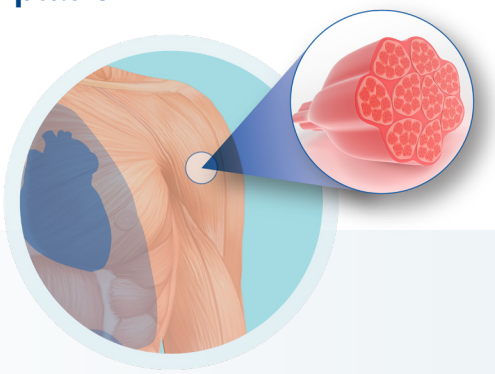
Targeting muscle with the goal of

stopping or reversing disease progression

Our FORCE™ platform

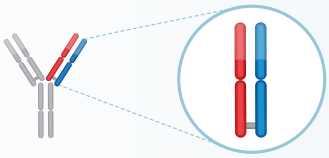


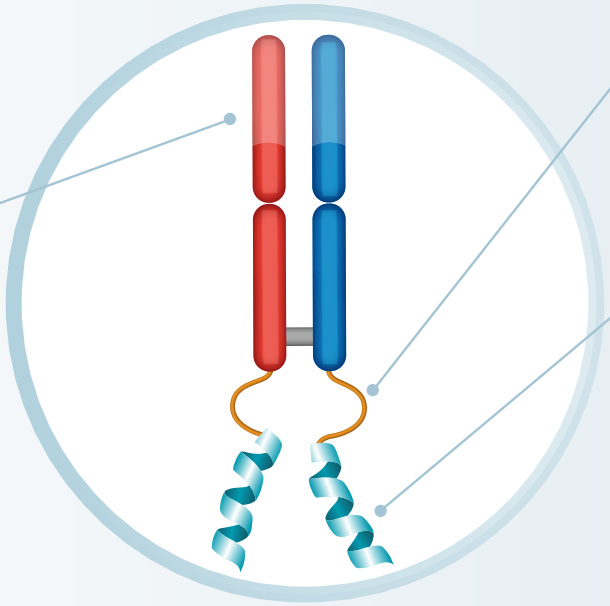
Our proprietary FORCE™ platform drives our efforts to develop targeted, modern oligonucleotide therapies that are designed to overcome limitations in delivery to muscle tissue seen with other approaches, with the goal of stopping or reversing progression of disease.

Our therapeutics consist of three components:

1. Antibody

Proprietary Fab targets TfR1 to enable muscle delivery



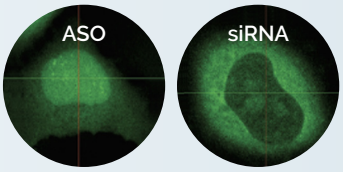


2. Linker

Clinically validated, enables precise conjugation of multiple payloads to a single Fab

3. Payload

Modularity enables rational selection of payload to target the genetic basis of disease



Nuclear localization

Cytoplasmic localization

Adapted from: Ohrt et al: *NAR*, 2006, v.34, p.1369

1. Our proprietary antigen-binding fragments (Fabs) are engineered to bind to a receptor which is highly expressed on muscle cells (TfR1) to enable targeted delivery to skeletal, cardiac and smooth muscle. We believe a Fab offers significant muscle-delivery advantages over a monoclonal antibody (mAb), including enhanced tissue penetration, increased tolerability and reduced risk of immune system activation.
2. We selected our linker based on the safety and efficacy it has demonstrated in approved products, as well as its stability and ability to release therapeutics within muscle cells.
3. Finally, we attach the Fab and linker to a therapeutic such as an antisense oligonucleotide (ASO), phosphorodiamidate morpholino oligomer (PMO), siRNA or small molecule.

©2021 Dyne Therapeutics, Inc.

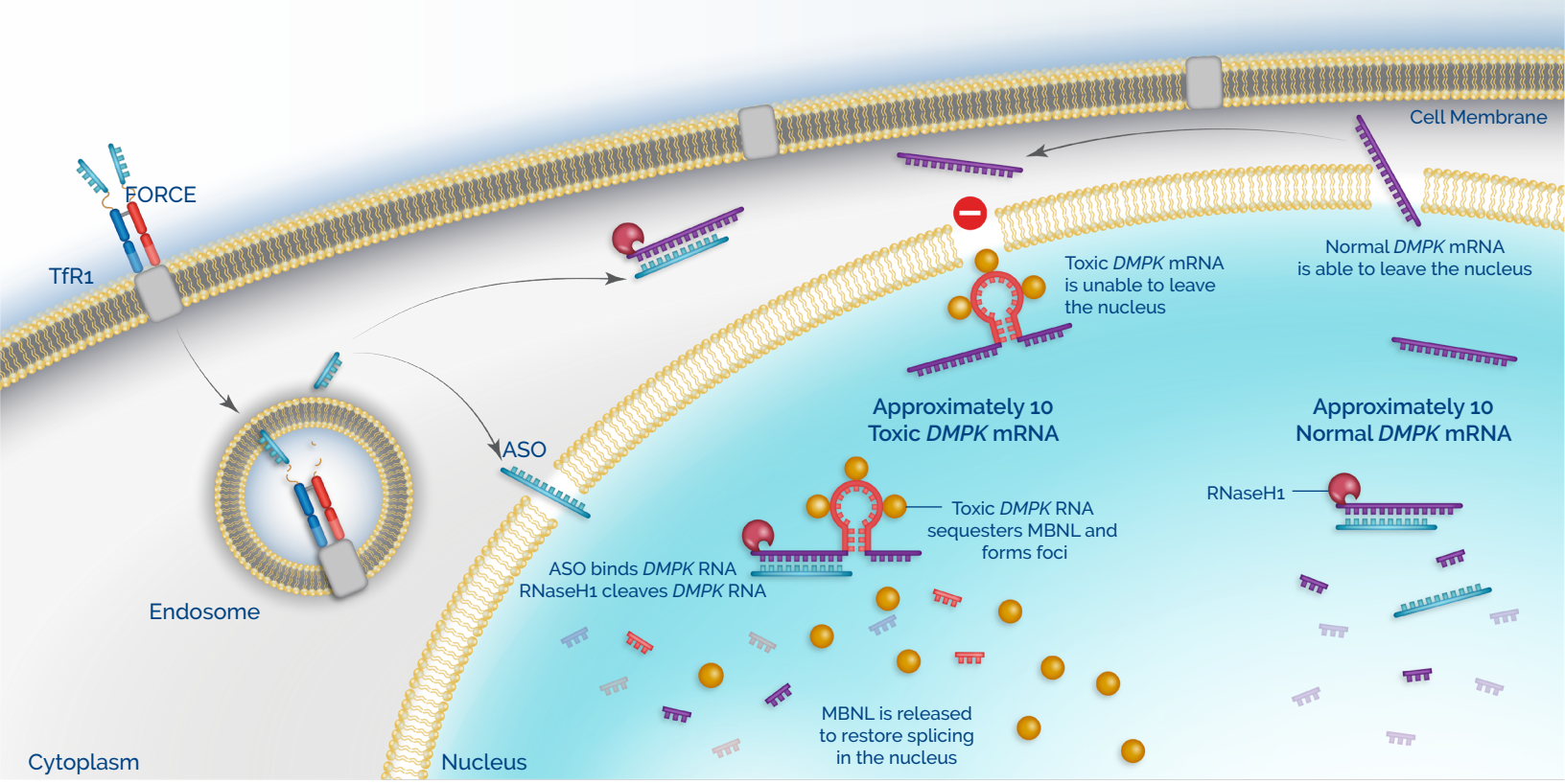
We believe our FORCE platform provides several advantages, including:

- Targeted delivery to muscle tissue
- Extended time between doses
- Ability to re-dose
- Ability to target the genetic basis of disease to stop/reverse progression

Our broad portfolio of therapeutic candidates for serious muscle diseases includes programs for myotonic dystrophy type 1 (DM1), Duchenne muscular dystrophy (DMD) and facioscapulohumeral muscular dystrophy (FSHD).

How FORCE targets DM1

FORCE addresses the genetic basis of disease in DM1 by targeting toxic nuclear RNA. DM1 is caused by a genetic defect introduced via abnormal expansion in a region of the *DMPK* gene. In a preclinical study, Dyne's lead DM1 candidate achieved robust knockdown of toxic human *DMPK* RNA in a novel mouse model that expresses the human TfR1 receptor and carries a human *DMPK* gene that represents a severe DM1 phenotype.



To learn more, visit

www.dyne-tx.com twitter linkedin facebook

©2021 Dyne Therapeutics, Inc.